Material Requirements Planning at A-CAT Corp

1. Case Diagnosis

The problem confronting Sunita Menon at A-CAT Corp extends well beyond meeting demand for the next eight weeks. With 240 units of P0110 already in stock and only 400 units required in Week 1, availability is not, in itself, a concern. The actual challenge lies in determining how to produce and order the required components economically, while accounting for a multi-level product structure, fixed lead times, and existing inventory. P0110 sits at the top of a three-level bill of materials: each unit requires two units of P0X and two units of P0Y, and each P0X, in turn, requires four units of P1X1 and three units of P1X2. None of these components carries independent demand; their requirements are derived entirely from the master schedule for P0110 and from one another. This is precisely the function that material requirements planning (MRP) performs. The master schedule establishes the weekly gross requirement for P0110; the bill of materials translates this into requirements for each component; lead times determine how far in advance orders must be released; and on-hand inventory is netted against gross requirements to arrive at the true quantity to be ordered. Because A-CAT applies fixed lot sizes rather than ordering to exact requirements, these net requirements are rounded upward, which is where unnecessary inventory build-up can arise.

2. MRP Matrix

Note: Week 1 shortfalls are shown as scheduled receipts (sized to the fixed lot quantity), since any order that would need to be released before Week 1 is assumed to be already committed. Projected available balance is shown at the start of each period.

Item: P0110 OH: 240 LT: 2 week(s) Lot size (Q): 613								
Period	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
Gross requirements	400	350	300	450	300	450	225	525
Scheduled receipts	613							
Projected available balance	**240**	**453**	**103**	**416**	**579**	**279**	**442**	**217**
Net requirements	160		197	34		171		308
Planned order receipts			613	613		613		613
Planned order releases	613	613		613		613		

Item: P0X OH: 500 LT: 1 week(s) Lot size (Q): 1226								
Period	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
Gross requirements	1226	1226		1226		1226		
Scheduled receipts	1226							
Projected available balance	**500**	**500**	**500**	**500**	**500**	**500**	**500**	**500**
Net requirements	726	726		726		726		
Planned order receipts		1226		1226		1226		
Planned order	1226		1226		1226			

releases								

Item: P0Y OH: 340 LT: 1 week(s) Lot size (Q): 1839								
Period	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
Gross requirements	1226	1226		1226		1226		
Scheduled receipts	1839							
Projected available balance	**340**	**953**	**1566**	**1566**	**340**	**340**	**953**	**953**
Net requirements	886	273				886		
Planned order receipts		1839				1839		
Planned order releases	1839				1839			

Item: P1X1 OH: 500 LT: 1 week(s) Lot size (Q): 2452								
Period	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
Gross requirements	4904		4904		4904			
Scheduled receipts	4904							
Projected available balance	**500**	**500**	**500**	**500**	**500**	**500**	**500**	**500**
Net requirements	4404		4404		4404			
Planned order receipts			4904		4904			
Planned order releases		4904		4904				

Item: P1X2 OH: 600 LT: 1 week(s) Lot size (Q): 1839								
Period	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
Gross requirements	3678		3678		3678			
Scheduled receipts	3678							
Projected available balance	**600**	**600**	**600**	**600**	**600**	**600**	**600**	**600**
Net requirements	3078		3078		3078			
Planned order receipts			3678		3678			
Planned order releases		3678		3678				

3. Short Interpretation

Treating the Week 1 shortfall as a scheduled receipt already in hand, rather than as a planned order released before the start of the horizon, allows every item to begin the plan with the necessary supply already available. This is a reasonable simplifying assumption, though it does not reveal how that initial batch was actually authorized. The item most worth flagging to Ratnaparkhi is P0Y: its fixed lot size of 1,839 units considerably exceeds the recurring requirement of 1,226 units, so a single order frequently covers more than one period's need, leaving inventory as high as 1,566 units unused for extended stretches of the plan. P1X1 and P1X2 exhibit a similar, though smaller, pattern. Before approving this

plan, Menon should question whether the large fixed lot sizes, presumably adopted to reduce the frequency of set-up costs, justify the additional carrying cost they generate, particularly given that A-CAT already entered this planning cycle with higher-than-normal inventory levels.